Privacy-Preserving Few-Shot UAV Identification on Embedded Hardware

Research Proposal — KAI Vision Platform

Prepared for: **Embention — Vision Department**

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Classification: Internal — Concept & Research Plan

This document proposes a research line for adding client-enrollable, privacy-preserving target identification to the KAI vision pipeline. It defines the operational concept, the technical approach, the privacy architecture, the experimental plan, a development roadmap, and the decision points where Embention input is required. Its purpose is to obtain approval to begin the research and prototyping phase.

1. Executive Summary

Embention's clients increasingly need ground- or air-based systems that can recognise a **specific** UAV — not merely detect that "a drone" is present. The operational concept is simple: a third-party client supplies a small set of reference images of a target UAV taken from different angles; the device memorises that target; and when the target appears in the sky at 50–100 m, the system identifies it and triggers the appropriate action (alerting, tracking, or guiding an interceptor towards it).

Two constraints make this non-trivial. First, **privacy**: the reference images belong to defence clients and must never leave the client's system — they cannot transit through Embention's infrastructure or any external server. Second, **edge deployment**: the system must run fully offline on KAI's embedded compute (TI TDA4VM-class SoC), under GPS denial, jamming, and degraded link conditions.

This proposal addresses both with a **few-shot metric-learning pipeline**: a generic detector (YOLO) finds flying objects, and a compact embedding network compares each detection against a mathematical signature of the target. Critically, the client enrols the target through a local application that runs the embedding model on **their own machine**; only an irreversible numerical gallery file (`gallery.npy`, ~kilobytes) is transferred to the device. Raw images never leave the client. The embedding model itself is trained entirely on public and synthetic UAV data, so no client data is ever needed for training.

A first feasibility study has already been carried out: the architecture has been selected and validated at design level, the embedded export path to the TDA4VM accelerator (TIDL) has been proven viable, and the optical geometry of the 50–100 m engagement envelope has been quantified. We now request approval to proceed with the full research and prototyping phase described in Section 8.

2. Objectives and Scope

Primary objective. Develop and validate a pipeline that, given N client-supplied images of a target UAV (N to be determined experimentally), identifies that UAV in live video at 50–100 m with quantified precision, recall, and latency on embedded hardware.

Secondary objectives.

(a) Determine experimentally the minimum number and type of enrollment images (angles, backgrounds, lighting) that the client must provide for reliable identification — this becomes the client-facing enrollment specification. (b) Guarantee, by construction, that client imagery cannot be reconstructed or exfiltrated from anything that leaves the client's premises. (c) Demonstrate that the identification model fits the KAI compute budget alongside the existing detection stage. (d) Characterise failure modes (distance, motion blur, low light, similar-looking decoys) and define confidence thresholds and fallback behaviour.

Out of scope for this phase. Final hardware integration and flight-control coupling (interception guidance) are deferred, as agreed; this phase concentrates on the embedding pipeline, the privacy architecture, and the enrollment specification. The analysis is not limited to the currently available sensors: the 1920×1080 / 30°–45° HFOV configuration constrains field testing, not the study itself.

3. Operational Concept

Enrollment (client side). The client installs a lightweight enrollment application on their own system. They load the reference images of the target UAV; the application runs the embedding model locally and produces a compact target signature. Only this signature is transferred to the KAI device — by file, by link, or physically. The images themselves are deleted or retained at the client's discretion; Embention never sees them.

Operation (device side). In flight or on the ground, the device runs its standard detection model (YOLO-class) over the camera stream. Every detection crop of a flying object is passed to the identification model, which converts it into the same numerical space as the enrolled signature and computes a match score. Above a calibrated confidence threshold, the target is declared identified and the configured action is triggered; below it, the object is treated as unknown. The whole loop runs offline on the device — no connectivity is required at any point during operation.

Engagement envelope. For a ~2 m × 1 m fixed-wing UAV observed with a 1920×1080 sensor and a 30°–45° HFOV lens, the target subtends approximately 46–143 pixels across the 50–100 m range. This pixel envelope is small but workable for embedding-based identification, and it directly drives the experimental design in Section 7: enrollment images are high quality, but operational crops are small, blurred and noisy, so the pipeline must be trained and evaluated across exactly this resolution gap.

4. Technical Approach

4.1 Why few-shot metric learning

A conventional classifier would need retraining for every new target — impossible when the client cannot share data and the device is offline. Instead we use **metric learning**: a network is trained once, on public and synthetic UAV datasets, to map images of the same aircraft close together in an embedding space and images of different aircraft far apart. Enrolling a new target then requires no training at all — only a forward pass over the client's reference images. This is the same family of techniques used in face recognition, adapted to the UAV domain.

4.2 Selected architecture

The feasibility study converged on the following configuration, chosen for the best balance of accuracy, embedded efficiency, and TIDL compatibility:

Component	Decision and rationale
Detection stage	Generic YOLO-class detector (UAV-only target class). The identification model receives only bounding-box crops, never full frames — the full image is not processed directly.
Identification model	Prototypical Network with a MobileNetV3-Small backbone — ~2.5 M parameters, designed for mobile/embedded inference.
Embedding	128-dimensional, L2-normalised vectors; matching by cosine/Euclidean distance against the enrolled prototype.
Prototype aggregation	Attention-weighted aggregation of the client's reference views, so the system adaptively emphasises the enrollment angles most similar to the current observation instead of naively averaging all views.
Enrollment degradation	A configurable pipeline (off by default) that optionally degrades enrollment images to the operational pixel envelope (46–143 px) to close the quality gap between client photos and in-flight crops; its benefit is one of the experimental questions.
Training data	Public UAV datasets, self-captured footage, and synthetic renders. No client data is used or needed for training.

4.3 Handling the enrollment-vs-operation image gap

This is the central scientific risk of the project: the client's photos (close-range, sharp, ground background) look nothing like an operational crop (50–100 m, tens of pixels, sky background, motion blur). We attack it from three directions simultaneously: training-time augmentation that simulates the operational degradation; the optional enrollment-side degradation pipeline above; and the attention-weighted prototype, which reduces sensitivity to mismatched viewpoints. The experiments in Section 7 quantify how much each contributes.

5. Privacy and Security Architecture

The governing requirement is absolute: **client images must never reach Embention or any external server.** The selected design (referred to internally as Option A) satisfies this by construction rather than by policy:

Property	How it is achieved
Local processing	The enrollment application ships the embedding model in ONNX form and runs it entirely on the client's machine. No network connection is required during enrollment.
Minimal artefact	The only file produced is the target signature (gallery.npy): a handful of 128-dimensional vectors, a few kilobytes. This is what travels to the device.
Irreversibility	Embeddings are a lossy, many-to-one compression. The original image cannot be reconstructed from a 128-dim vector; at most, the signature confirms similarity to an image one already possesses. We will document this property formally in the deliverables, including an inversion-attack analysis.
No training dependency	Because the model is trained exclusively on public/synthetic data, there is no federated-learning round, no gradient sharing, and no data pooling — entire categories of leakage risk are eliminated rather than mitigated.
Offline device	During operation the device needs no connectivity; signatures are loaded once. This aligns with GPS-denied and jammed environments by default.
Hardening (later phase)	Signature-file integrity (signing), encrypted storage on device, and tamper logging are planned for the productisation phase; they do not affect the research-phase architecture.

Compared with the alternatives considered in the feasibility study — federated learning (complex, still leaks gradients, unnecessary here), secure enclaves (hardware-dependent, premature), or sending images for server-side processing (violates the requirement outright) — Option A is the only approach that is simultaneously simple, auditable, and provably compliant with the data-residency constraint.

6. Embedded Deployment Strategy

The target compute class is the TI TDA4VM / AM69A family used by KAI (SDK Linux 11-01-00-03, Baremetal 11_01_00_04). The feasibility study already resolved the main compilation obstacle: the TIDL graph compiler does not support the LayerNorm operator used in the projection head. The model is therefore exported in two parts:

Artefact	Content and execution target
backbone_tidl.onnx	MobileNetV3-Small features + average pooling (output 1×576×1×1). Fully supported by TIDL and accelerated on the C7x/MMA deep-learning accelerator — this is >95% of the compute.
projection_head.onnx	Flatten + Linear + LayerNorm + L2 normalisation (output 1×128). Runs on the ARM cores via ONNX Runtime; negligible cost (a single small matrix multiply).

This split keeps the accelerated graph clean for INT8 quantisation while preserving exact numerical behaviour of the normalisation stages on CPU. The identification model only processes detector crops (not full frames), so its per-frame cost scales with the number of airborne objects — typically one or two — leaving the FPS budget dominated by the detector, as in the current KAI pipeline. Final quantisation strategy (INT8 calibration set, per-layer fallback) and the FPS-vs-model-size trade-off will be settled on real hardware in Phase 3, per the agreed plan that hardware adaptation comes after the pipeline is validated.

7. Research Questions and Experimental Plan

The research phase is organised around the questions Embention and the client actually need answered:

Question	Experiment	Decision it informs
Q1 — Enrollment size	How many reference images are needed? Sweep $N \in \{1, 3, 5, 10, 20\}$ per target; measure identification accuracy vs N.	Recommended N and marginal value of each extra image.
Q2 — Enrollment content	Which views matter? Compare angle coverage strategies (orbit vs frontal-only vs random) and background/lighting diversity.	A concrete enrollment guide for clients (the 'what to photograph' spec).
Q3 — Quality gap	Does degrading enrollment images to the operational pixel envelope (46–143 px) help or hurt? Ablate the degradation pipeline on/off.	Default configuration of the enrollment app.
Q4 — Aggregation	Attention-weighted prototype vs simple mean of views.	Validated aggregation method (or simplification if attention adds nothing).
Q5 — Discrimination	Can the system distinguish the target from visually similar UAVs of the same class? Build a confusion set of look-alike airframes.	False-positive characterisation; confidence threshold calibration.
Q6 — Robustness	Performance vs distance (pixel size), motion blur, low light, partial occlusion, sky/cloud backgrounds.	Operating envelope definition and fallback thresholds.
Q7 — Embedded cost	Accuracy retention after INT8 quantisation; per-crop latency on TDA4VM.	Go/no-go for the selected backbone; FPS budget.

Evaluation protocol: episodic few-shot evaluation on held-out UAV identities (the model never sees test airframes during training), reporting precision/recall at the calibrated threshold, ROC-AUC for target-vs-impostor discrimination, and rank-1 identification accuracy. Metrics are reported per pixel-size bucket so performance maps directly onto distance.

8. Development Roadmap

Phase	Content	Indicative timing
Phase 0 — Feasibility (done)	Architecture selection, privacy design, optical/geometry analysis, TIDL export path proven (split backbone/head compiles).	Completed

Phase	Content	Indicative timing
Phase 1 — Training pipeline	Dataset assembly (public + synthetic), episodic training of the prototypical network, baseline accuracy on held-out identities. Q1–Q4 experiments.	Weeks 1–5
Phase 2 — Robustness & thresholds	Q5–Q6 experiments; confidence calibration; failure-mode catalogue; enrollment specification document for clients.	Weeks 5–8
Phase 3 — Embedded validation	INT8 quantisation on TIDL, on-target latency/accuracy measurement (Q7), integration of the two-stage detect→identify loop on the dev board.	Weeks 8–11
Phase 4 — Enrollment app prototype	Client-side application: local ONNX inference, gallery generation, packaging. End-to-end demo: client enrols → device identifies live.	Weeks 11–14
Phase 5 — Reporting & handover	Final results, privacy analysis (embedding irreversibility), recommendations for productisation (hardening, sensor options, interception coupling).	Weeks 14–16

MVP definition (end of Phase 4): a working demonstration in which a 'client' enrols a target UAV from N photographs on a separate machine, transfers only the signature file, and the device identifies that UAV in video at representative pixel sizes, in real time on TDA4VM-class hardware, with documented accuracy and latency. This is the artefact that supports a productisation decision.

9. Risks and Mitigations

Risk	Severity	Mitigation
Enrollment/operation domain gap too large (sharp photos vs 46–143 px crops)	High	Three independent mitigations already designed: degradation-matched training augmentation, optional enrollment degradation, attention-weighted prototypes. Q3/Q6 quantify residual risk early (Phase 1–2).
Look-alike UAVs cause false positives	High	Hard-negative mining with same-class airframes during training; impostor evaluation set (Q5); conservative threshold + 'unknown' fallback class; multi-frame temporal voting before declaration.
Public/synthetic training data insufficient for the UAV domain	Medium	Synthetic rendering pipeline to generate arbitrary airframes/viewpoints; self-captured footage; progressive dataset expansion is budgeted in Phase 1.
INT8 quantisation degrades embedding quality	Medium	Quantisation-aware fine-tuning if post-training quantisation underperforms; the projection head stays in float on ARM, protecting the final metric space.
Detector misses small/distant targets, starving the identifier	Medium	Detector and identifier are decoupled: detector recall can be tuned independently (lower threshold + identifier as verifier). Tiling/overlapping-patch strategy reserved as fallback.
Adversarial deception (painted decoys, visual spoofing)	Low (this phase)	Documented as a known limitation; multi-frame consistency and trajectory plausibility checks recommended for productisation; out of scope for the research phase.
Hardware/SDK changes during the project	Low	Pipeline is ONNX-first and hardware-agnostic by design; TIDL specifics are isolated in the export stage, per the agreement that hardware adaptation comes later.

10. Open Questions for Embention / Client

Approval of this proposal does not require answering these, but they shape Phases 2–4 and we would like to schedule them:

Operational. Should identification require multi-frame confirmation before triggering an action, and what reaction latency is acceptable? Is there a preference for false-negative vs false-positive risk (a missed target vs engaging the wrong object)? Will multiple targets need to be enrolled simultaneously, and up to how many?

Enrollment. What imagery can clients realistically provide — manufacturer photos, ground-truth flight footage, or only a handful of stills? May the enrollment app suggest retakes (quality feedback) or must it be fully passive? Who manages signature versioning when a target is re-enrolled?

Integration. Which exact camera/lens pairs are candidates beyond the current 1920×1080 / 30°–45° test configuration? What share of the KAI compute budget is available to the identification stage alongside the existing detector? What logging is acceptable on-device in operational environments (for post-mission review vs OPSEC constraints)?

11. Deliverables and Requested Decision

Deliverables of the research phase: (1) the trained identification model and training pipeline; (2) the dual ONNX export (TIDL backbone + ARM projection head) with embedded benchmark results; (3) the client enrollment application prototype; (4) the enrollment specification ('how many images, of what kind') backed by the Q1–Q3 experiments; (5) a privacy analysis documenting why the transferred signature cannot leak client imagery; (6) the final technical report with the robustness envelope, thresholds, and productisation recommendations. The work is additionally documented as a Bachelor's Thesis (TFG) at the University of Alicante, with Embention review of any published material.

Requested decision. We ask Embention to approve the start of Phase 1 (training pipeline and enrollment experiments) on the architecture summarised above. No client data, no hardware changes, and no external services are required to begin: the phase runs entirely on public and synthetic data and existing development hardware.

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